2: Compute $\frac{\partial C}{\partial a_i}$

$$C = -\sum_{i=1}^{m} y_i \ln a_i$$

$$\frac{\partial C}{\partial a_i} = -\frac{y_i}{a_i} \qquad \nabla_a C = \begin{bmatrix} -\frac{y_1}{a_1} \\ -\frac{y_2}{a_2} \\ \vdots \\ -\frac{y_m}{a_m} \end{bmatrix}$$

$$a_j = \frac{e^{s_j}}{\sum_{m=1}^{m} e^{s_m}} \qquad \frac{\partial a_i}{\partial s_j} = a_i(\delta_{ij} - a_j)$$

where $\delta_{ij} = 1$ if $i = j$ and otherwise

$$A = \left[\frac{\partial a_i}{\partial s_j}\right]$$

$$\frac{\partial a_i}{\partial s_j} = a_i(\delta_{ij} - a_j)$$

$$\delta_j = \frac{\partial C}{\partial s_j}$$

$$\delta_j = \frac{\partial C}{\partial s_j} = \sum_{i=1}^{M} \frac{\partial C}{\partial a_i}\frac{\partial a_i}{\partial s_j}$$

$$\delta_j = \sum_{i=1}^{M} \left(-\frac{y_i}{a_i}\right) a_i(\delta_{ij} - a_j)$$

$$= -\sum_{i=1}^{M} y_i \delta_{ij} + \sum_{i=1}^{M} y_i a_j$$

$$\delta_j = -y_j + a_j \sum_{i=1}^{M} y_i$$

$$\sum_{i=1}^{M} y_i = 1$$

$$\delta = \begin{bmatrix} a_1 - y_1 \\ a_2 - y_2 \\ a_M - y_M \end{bmatrix} = a - y$$

$$\delta = \nabla_s C = a - y$$